INCH-POUND
MIL-DTL-18030F(SH)
17 April 2018
SUPERSEDING
MIL-V-18030E(SH)
24 August 1990

DETAIL SPECIFICATION

# VALVES, CONTROL, AIR-DIAPHRAGM-OPERATED (COMPLETE WITH PNEUMATIC INSTRUMENTATION)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers air-diaphragm-operated control valves complete with their associated pneumatic instrumentation.

1.2 Classification. Control valves are of the following types and compositions as specified (see 6.2).

1.2.1 Types.

a. Type I – Single-seated, unbalanced.
b. Type II – Single-seated, direct piston balanced.
c. Type III – Single-seated, piston balanced incorporating an equalizing valve.
d. Type IV – Double-seated, semi-balanced.
e. Type V – Three-way, converging service (two inlets – one outlet).
f. Type VI – Three-way, diverging service (one inlet – two outlets).

1.2.2 Composition.

a. Composition A – 2¼ percent chromium – 1 percent molybdenum.
b. Composition B – 1¼ percent chromium – ½ percent molybdenum.
c. Composition D – Carbon steel.
d. Non-ferrous.

1.3 Ratings. Ratings are as specified in 3.3.2.1.1 and 3.3.2.1.2 (see 6.2).

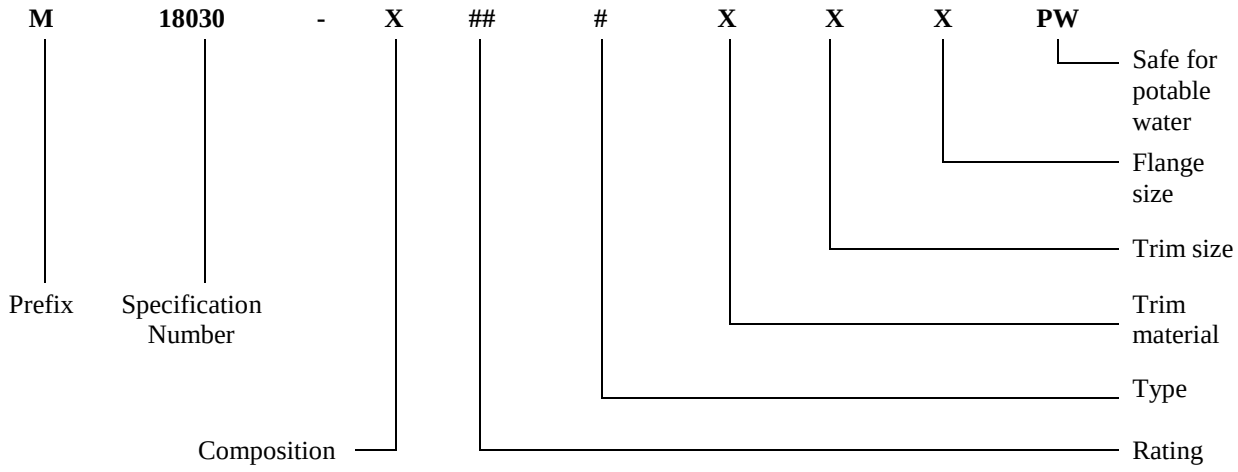
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1.4 Part or identifying number (PIN). PINs are for general information only. The acquisition requirements (see 6.2) take precedence. The PINs to be used for valves acquired to this specification are created as follows:




PIN Code:

| Composition Code | | Rating Code | | Type Code | | Trim Material Code | |
|---|---|---|---|---|---|---|---|
| **Composition** | **Code** | **Rating (pressure, psi)** | **Code** | **Type** | **Code** | **Trim material (see 3.2.1)** | **Code** |
| Composition A | A | Class 150 | 01 | I | 1 | Cobalt alloy | A |
| Composition B | B | Class 600 | 06 | II | 2 | Hardened CRES | B |
| Composition D | D | Class 900 | 09 | III | 3 | 18-8 CRES | C |
| Non-ferrous | N | Class 1500 | 15 | IV | 4 | Nickel-copper alloy | D |
| | | | | V | 5 | | |
| | | | | VI | 6 | | |



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| Trim and Flange Size Code | | Potable Water Use Code | |
|---|---|---|---|
| **Trim and flange size (inches nps)** | **Code** | **Potable water use** | **Code** |
| 0.5 | A | Safe for potable use | PW |
| 0.75 | B | Not safe for potable use | (blank) |
| 1.0 | C | | |
| 1.25 | D | | |
| 1.5 | E | | |
| 2.0 | F | | |
| 2.5 | G | | |
| 3.0 | H | | |
| 3.5 | J | | |
| 4.0 | K | | |
| 5.0 | L | | |
| 6.0 | M | | |
| 8.0 | N | | |
| 10.0 | P | | |
| 12.0 | R | | |

Example:
M18030-B095CHJPW - Air-diaphragm-operated control valve in accordance with MIL-DTL-18030, made from 1¼ percent chromium - ½ percent molybdenum, class 900, 1000 °F max, three-way, converging service (two inlets - one outlet) design, 300 series CRES plug and seat ring, 17-4Ph or 400 series CRES guide bushings or cage, 3-inch trim size and 3.5-inch flanges, potable water safe.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

QQ-N-286 - Nickel-Copper Aluminum Alloy, Wrought (UNS N05500)



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DEPARTMENT OF DEFENSE SPECIFICATIONS

| | | |
|---|---|---|
| MIL-DTL-901 | - | Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment, and Systems, Requirements for |
| MIL-DTL-1222 | - | Studs, Bolts, Screws, and Nuts for Applications Where a High Degree of Reliability is Required; General Specification for |
| MIL-DTL-15024 | - | Plates, Tags, and Bands for Identification of Equipment, General Specification for |
| MIL-R-17131 | - | Rods and Powders, Welding, Surfacing |
| MIL-PRF-20042 | - | Flanges, Pipe, Bronze (Silver Brazing) |
| MIL-G-24716 | - | Gaskets, Metallic-Flexible Graphite, Spiral Wound |

DEPARTMENT OF DEFENSE STANDARDS

| | | |
|---|---|---|
| MIL-STD-167-1 | - | Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited) |
| MIL-STD-798 | - | Nondestructive Testing, Welding, Quality Control, Material Control and Identification and HI-Shock Test Requirements for Piping System Components for Naval Shipboard Use |

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.GSAglobalsupply.gsa.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

| | | |
|---|---|---|
| S9074-AR-GIB-010/278 | - | Requirements for Fabrication, Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels |

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ [by searching for the document number without the suffix]. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/ AMERICAN SOCIETY FOR QUALITY (ASQ)

| | | |
|---|---|---|
| ANSI/ASQ Z1.4 | - | Sampling Procedures and Tables for Inspection by Attributes |

(Copies of this document are available online at http://webstore.ansi.org/ or http://asq.org/quality-press/index.html.)



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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ANSI/ASME B1.1 - Unified Inch Screw Threads (UN and UNR Thread Form)

(Copies of this document are available online at http://webstore.ansi.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.12 - Class 5 Interference-Fit Thread

ASME B16.5 - Pipe Flanges and Flanged Fittings

(Copies of these documents are available online at http://webstore.ansi.org.)

ASTM INTERNATIONAL

ASTM A105/A105M - Standard Specification for Carbon Steel Forgings for Piping Applications

ASTM A182/A182M - Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service

ASTM A216/A216M - Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service

ASTM A217/A217M - Standard Specification for Steel Castings, Martensitic Stainless and Alloy, for Pressure-Containing Parts, Suitable for High-Temperature Service

ASTM A276/A276M - Standard Specification for Stainless Steel Bars and Shapes

ASTM A395/A395M - Standard Specification for Ferritic Ductile Iron Pressure-Retaining Castings for Use at Elevated Temperatures

ASTM A479/A479M - Standard Specification for Stainless Steel Bars and Shapes for Use in Boilers and Other Pressure Vessels

ASTM A494/A494M - Standard Specification for Castings, Nickel and Nickel Alloy

ASTM A564/A564M - Standard Specification for Hot-Rolled and Cold-Finished Age-Hardening Stainless and Heat-Resisting Steel Bars and Shapes

ASTM A675/A675M - Standard Specification for Steel Bars, Carbon, Hot-Wrought, Special Quality, Mechanical Properties

ASTM B26/B26M - Standard Specification for Aluminum-Alloy Sand Castings

ASTM B61 - Standard Specification for Steam or Valve Bronze Castings

ASTM B369 - Standard Specification for Copper-Nickel Alloy Castings

(Copies of these documents are available online at www.astm.org.)

NSF INTERNATIONAL

NSF/ANSI 61 - Drinking Water System Components - Health Effects

(Copies of this document are available online at www.nsf.org.)



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SAE INTERNATIONAL

SAE J1086 - Numbering Metals and Alloys

SAE J1926/1 - Connections for General Use and Fluid Power - Ports and Stud Ends with ASME B1.1 Threads and O-Ring Sealing - Part 1: Threaded Port with O-Ring Seal in Truncated Housing

SAE AMS5373 - Alloy Castings, Sand, Corrosion and Heat Resistant 16Co – 29Cr – 4.5W, As Cast

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials shall be as specified in table I. Materials for parts other than those listed in table I shall be selected to prevent galling, seizing, or excessive wear on operating parts. Clearance shall prevent interferences due to thermal expansion. All materials that may come in contact with potable freshwater shall be NSF/ANSI 61 approved for drinking water contact (see 6.2).

TABLE I. List of materials.

| Name of part | Composition A | Composition B | Composition D | Non-ferrous 1/ |
|---|---|---|---|---|
| Body bonnet and bottom flange | ASTM A182/A182M, grade F22<br>ASTM A217/A217M, grade WC9 | ASTM A182/A182M, grade F11<br>ASTM A217/A217M, grade WC6 | ASTM A105/A105M,<br>ASTM A216/A216M, grade WCB<br>ASTM A675/A675M, grade 60 | ASTM B61<br>ASTM B369, alloy 964 |
| Yoke | ASTM A395/A395M<br>ASTM A216/A216M, grade WCB<br>ASTM B26/B26M | | | |
| Gaskets | Steam service – all pressure ratings<br>Oil and water service – gauge pressure rating 600 lb/in$^2$ and above – MIL-G-24716 | | | |
| Bolting 2/ | | | | |
| Externally threaded | Grade B16 | Grade B16 | Grade B7 | Grade B7 |
| Internally threaded | Grade 7 | Grade 7 | Grade 2H | Grade 2H, 400, or 405 |
| Packing | See 3.3.2.8 | | | |
| Diaphragm | See 3.3.3.5 | | | |
| Internal trim | See 3.2.1 | | | |



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TABLE I. List of materials – continued.

| Name of part | Composition A | Composition B | Composition D | Non-ferrous [1/] |
|---|---|---|---|---|
| Stem and packing box | ASTM A276/A276M or ASTM A479/A479M | | | Nickel-Copper, QQ-N-281 Austenitic CRES, ASTM A276/A276M or ASTM A479/A479M ASTM B61 |
| Piston | Martensitic or precipitation hardening CRES-hardened, ASTM A564/A564M, type 630, or ASTM A276/A276M or ASTM A479/A479M | | | Nickel-Copper, QQ-N-281 Austenitic CRES, ASTM A276/A276M or ASTM A479/A479M ASTM B61 |
| Other parts | See 3.2.1 | | | |

NOTES:

[1/] Where non-ferrous valves are to be installed in sea water services, all internal trim and external hardware (i.e., bolting and packing glands) shall be of nickel-copper alloys.

[2/] All bolting shall be in accordance with MIL-DTL-1222. Higher temperature rated bolting materials may be used in lower temperature applications.

3.2.1 Internal trim materials. Unless otherwise specified (see 6.2), materials for the seat or seat ring, plug, guide posts, bushings (if not cage guided), and cage, where applicable, shall consist of the materials specified in 3.2.1.a through 3.2.1.d.

a. Trim A, cobalt alloy. Seating and guiding surfaces shall be either wrought or cast cobalt alloy, or an inlay of cobalt alloy not less than 3/32 inch thick for the main seat and disc surfaces. Where inlays are used, welding rods shall be as listed below.

b. Trim B, hardened CRES. Plugs shall be hardened 400 series or 17-4Ph CRES. Seat rings shall be cobalt alloy. Guiding surfaces shall be hardened CRES or cobalt alloy.

c. Trim C, 18-8 CRES. Plugs and seat rings shall be 18-8 (300 series) CRES. Guide bushings or cage shall be hardened 17-4Ph or 400 series CRES.

d. Trim D, nickel-copper alloy. All trim components shall be monel, K-monel, or S-monel.

Trim materials for 3.2.1.a through 3.2.1.d above shall be in accordance with table II.

TABLE II. Trim materials.

| | |
|---|---|
| Wrought or case cobalt alloy | SAE AMS5373 |
| Inlaid cobalt alloy (welding rod) | MIL-R-17131, type MIL-RCo-Cr-A |
| Hardened 400 series CRES | ASTM A276/A276M or A479/A479M |
| 18-8 series CRES | ASTM A276/A276M or A479/A479M |
| 17-4Ph CRES | ASTM A564/A564M, type 630 |
| Monel | QQ-N-281, UNS N04400 |



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TABLE II. Trim materials – continued.

| | |
|---|---|
| K-monel | QQ-N-286 in the annealed age-hardened condition, UNS N05500 |
| S-monel | ASTM A494/494M, UNS N24025 |
| NOTE:<br>1. Unified numbering system (UNS) numbers are as defined in SAE J1086. | |

3.2.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Construction. Construction shall be as specified (see 6.2), to include composition, rating, type, trim material, trim and flange size, and potable water usability, and as specified in 3.3.1 through 3.3.4.

3.3.1 General requirements.

3.3.1.1 Threads. Threads shall be in accordance with ANSI/ASME B1.1. Where necessary, provisions shall be incorporated to prevent the accidental loosening of threaded parts. Pipe threads shall not be used.

3.3.1.2 Bolts and nuts. Hex head bolts and nuts shall be in accordance with MIL-DTL-1222.

3.3.1.3 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.3.1.4 Accessibility. Valves shall be constructed to be accessible for adjustment and repair without requiring removal from the line.

3.3.1.5 Maintainability. Internal parts shall be constructed to permit easy disassembly and reassembly with standard tools or special tools and to prevent the incorrect reassembly of the valve. Positioning and alignment of all parts in the assembly shall employ positive means so that correct reassembly is repeatedly assured. Special tools required for maintenance of the valve shall be as specified (see 6.2). Special tools are defined as those tools not listed in the GSA Global Supply Catalog.

3.3.2 Body subassembly.

3.3.2.1 Pressure-temperature ratings.

3.3.2.1.1 Composition A, B, and D valves. The design and pressure-temperature ratings for composition A, B, and D valves shall be in accordance with ASME B16.5 for classes 150, 600, 900, and 1500, except for the maximum temperature limitations as follows:

Composition A – 1050 °F

Composition B – 1000 °F

Composition D – 775 °F

3.3.2.1.2 Non-ferrous valves. Non-ferrous valves shall be constructed for a working gauge pressure of 100 pounds per square inch ($lb/in^2$) at 425 °F.

3.3.2.2 End preparation. Valves shall be furnished with flanged ends in accordance with ASME B16.5 for composition A, B, and D valves and MIL-PRF-20042 for non-ferrous valves. Flanges shall be cast or forged integral with the valve body. The inlet and outlet flange shall be of the same size and pressure rating. Unless otherwise specified (see 6.2), valves shall be of a basic globe configuration with inline inlet and outlet ports.



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3.3.2.3 Body passages. For control valves intended for application under erosive conditions, body passages shall be constructed to produce gradual changes in flow direction so as to reduce any effects of concentrated impingement and 90-degree turns.

3.3.2.4 Bonnet and bottom flange joints.

3.3.2.4.1 Composition A, B, and D valves. Bonnet and bottom flanges, where applicable, for composition A, B, and D valves shall be located by body guiding for attachment to the body, a close tolerance fit between machined diameters on the body, bonnet, and bottom flange, rather than depending on studs or bolts for location. Where spiral wound gaskets are used, they shall be fully retained and the joints shall have metal-to-metal take-up to provide controlled compression of the gaskets. Flanges shall be secured by either of the following:

a. Through bolts threaded the entire length and fitted with a nut on each end. Threads on bolts and nuts shall be a class 2 fit in accordance with ANSI/ASME B1.1.

b. Studs shall be in accordance with MIL-DTL-1222, style b, except the dimensions of the interference fit shall be in accordance with ASME B1.12.

3.3.2.4.2 Non-ferrous valves. For non-ferrous valves in sizes 1½ inches and below, a straight machine threaded and gasketed bottom cap may be utilized and, in non-ferrous valves of all sizes, the bonnet and bottom flange joints may be secured with cap screws in addition to the methods specified in 3.3.2.4.1.a and 3.3.2.4.1.b. Joint construction shall assure parallel alignment of the valve guiding surfaces.

3.3.2.5 Spot or back facing. Bearing surfaces of nuts and their respective surfaces on the valves shall be finish machined on composition A, B, and D valves and cast smooth and true without nut interference, or be finish machined for non-ferrous valves.

3.3.2.6 Bonnet construction. Valves shall be constructed to limit packing temperatures to a maximum of 500 °F under any operating condition for which the valve is designed. Where necessary, a radiation or extension bonnet shall be provided for this purpose. A packing lubricator shall not be used.

3.3.2.7 Packing box assembly. A packing box assembly shall be provided to seal against leakage along the stem. Sealing bellows shall not be used. The packing gland shall be of bolted construction and shall be constructed so that the actuator assembly can be removed without disturbing the packing adjustment.

3.3.2.8 Packing. Packing shall be designed for sustained operation at the temperature and pressure conditions to which it will be subjected.

3.3.2.9 Gasket. The spiral wound gasket shall retain sufficient residual load in service to maintain a leak-tight joint over at least a 3-year period under normal operating conditions.

3.3.2.10 Internal trim.

3.3.2.10.1 Guiding surfaces. A difference in hardness of at least 100 points Brinell shall be maintained between the rubbing surfaces for main valve guiding. This requirement does not apply if both the guide posts and bushings are stellited or if the hardness of either exceeds 450 Brinell. Rubbing surfaces for valve guiding shall have a finish of roughness height rating (RHR) of 16 or better.

3.3.2.10.2 Seat rings. Where quick change cage trim is not required (see 6.2), threaded-in seat rings may be provided. On valves used on steam service greater than a gauge pressure of 225 $lb/in^2$, seat rings shall be either seal welded circumferentially or gasketed. On other applications of threaded-in seat rings, a seat may be provided by a make-up against machined surfaces. In no case shall close fitting threads be used to provide a seal. Threaded-in seat rings shall be provided with substantial lugs to facilitate installation and removal.

3.3.2.11 Coupled valve stem. The valve stem and valve plug/disc shall be coupled together.



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3.3.3 Diaphragm actuator assembly.

3.3.3.1 Operating range. Unless otherwise approved by NAVSEA, the actuator assembly shall operate on a standard gauge pressure of 3 to 15 or 6 to 30 lb/in$^2$ air signal ranges.

3.3.3.2 Reversibility. Direct acting and reverse acting actuators shall be interchangeable on body subassemblies without requiring modification or additional parts to the body subassemblies.

3.3.3.3 Yoke. Yoke construction shall allow easy access to the stuffing box, stem connection, and spring adjuster from either side of the valve. Universal mounting pads shall be provided on opposite sides of the yoke for mounting valve positions or other accessories, or both.

3.3.3.4 Travel indicator. A travel indicator shall be provided to indicate valve plug position. The indicator scale shall be attached to the yoke and marked in inches on one side of a dividing line and the percentage of opening on the other.

3.3.3.5 Diaphragms. Diaphragms shall be deep molded and uniform contact shall be maintained between the diaphragm plate and diaphragm during full valve travel. Diaphragms shall be molded from neoprene or Buna-N and shall be synthetic fabric reinforced to provide a minimum burst strength of 150 lb/in$^2$ over a 2-inch opening.

3.3.3.6 Diaphragm spring. When practicable, only one diaphragm spring shall be required. The ends of the spring shall be ground within 1 degree of its perpendicular axis. The spring compression shall be readily adjustable while the valve is in service. Spring and spring parts shall be coated with a rust inhibiting material.

3.3.3.7 Diaphragm case. The diaphragm case shall be constructed for a maximum operating pressure of 50 lb/in$^2$ gauge. The case shall be made of steel coated with a rust-inhibiting material, or aluminum.

3.3.3.8 Air connections. Air connections between the pilot, actuator, and other accessories shall be in accordance with SAE J1926/1, straight thread and O-ring seal.

3.3.3.9 Manual override. When specified (see 6.2), a hand wheel for manual override shall be furnished. The location shall be as specified (see 6.2). A clockwise rotation of the hand wheel shall close the valve.

3.3.4 Instrumentation. Unless otherwise specified (see 6.2), the pneumatic instrumentation necessary to accomplish the required control function (see 6.3.8) shall be furnished as part of the valve. The process connection to controllers, pilots, or transmitters shall be flanged. The overall construction shall permit the use of the most simple and direct instrumentation system consistent with performance requirements. Construction of all instrumentation shall emphasize simplicity, maintainability, ruggedness, and reliability. Wherever compatible from a performance, sensitivity, and speed of response standpoint, intermittent bleed instrumentation (see 6.3.9) shall be used.

3.4 Performance.

3.4.1 Capacity. The required capacity of the valve shall be as specified (liquid [gallons per liter per minute] or gas [cubic feet per liter per minute]) (see 6.2).

3.4.2 Rangeability. Rangeability (see 6.3.12) for the various flow characteristics shall be as follows:

a. Equal percentage – 50-to-1, minimum

b. Modified linear – 35-to-1, minimum

c. Linear – 25-to-1, minimum

d. Quick opening – 15-to-1, minimum



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3.4.3 Seat tightness. Maximum allowable leakage, under the test conditions specified in 4.3.4, shall not exceed the following values:

| Type | Percent of maximum valve capacity |
|---|---|
| I | 0.01 |
| II | 0.25 |
| III | 0.01 |
| IV | 0.50 |
| V | 0.01 (in each direction) |
| VI | 0.01 (in each direction) |

3.4.4 External leakage, valve body subassembly. There shall be no visible external leakage.

3.4.5 Leakage, diaphragm case. There shall be no leakage past the diaphragm or externally from the diaphragm case (see 4.3.5).

3.4.6 Flow characteristics. The inherent flow characteristics ($C_v$) for control valves covered by this specification shall be as specified (see 6.2) as per the following:

a. Equal percentage. The percentage change in $C_v$ per increment of lift shall remain constant over the entire operating range of the valve. This characteristic is based on a minimum $C_v$, with plug just off the seat, of 2 percent maximum capacity. The ideal characteristic is an 8⅓-percent increase in lift producing a 38-percent increase in $C_v$. The actual achieved increase in $C_v$ shall be from 33 to 43 percent.

b. Linear. The change in $C_v$ increment of lift shall remain constant over the entire operating range of the valve. Actual characteristics may vary ±5 percent, for linearity, if minimum $C_v$, with plug just off the seat, does not exceed 4 percent of maximum capacity.

c. Modified linear (see 6.3.6). Minimum $C_v$, with plug just off the seat, shall not exceed 3 percent of maximum capacity.

d. Quick opening. $C_v$ shall increase rapidly with lift and reach a maximum value at a lift equal to approximately 25 percent of the seat diameter.

e. Three-way. Total flow area shall remain constant at all positions of the valve stem.

3.4.7 Mechanical shock and vibration. Valves shall be constructed to meet the mechanical shock requirements specified in MIL-DTL-901, grade A, class I, and MIL-STD-798, and the environmental vibration requirements specified in MIL-STD-167-1, type I. This shall apply to the entire valve, including all associated instrumentation.

3.4.8 Leakage, stem. There shall be no visible leakage past the packing under operating conditions. For steam service, detection may consist of viewing against a dark background or use of a mirrored surface.

3.4.9 Hysteresis. Under operating conditions, and at any position within the full stroke of the stem, a change in signal pressure of 0.35 lb/in$^2$ in either direction shall cause valve movement. The maximum difference in valve position between increasing and decreasing pressure shall not exceed 2 percent of stroke.

3.4.10 Linearity. The relationship between air signal and stem travel shall be linear within ±3 percent.

3.5 Marking.

3.5.1 Body markings. The manufacturer’s name or trademark and the body material composition shall be cast or forged integral with the valve body. The size, rating, and a flow arrow or “inlet” and “outlet” shall be cast or forged integral with the valve body or stamped on the outside diameter (od) of the flanges.



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3.5.2 Identification plates. Each valve and pilot shall have an identification plate made of CRES or brass in accordance with MIL-DTL-15024. Identification plates shall be permanently fastened to a part of the valve not subjected to working pressure and which will not be covered by insulation. Identification plates shall contain the following information or a space therefor:

a. Manufacturer's Commercial and Government Entity (CAGE) code.
b. PIN (see 1.4).
c. Maximum inlet temperature, pressure rating, and pressure differential operating limitation.
d. Flow characteristics and flow coefficient ($C_v$) rating.
e. Diaphragm actuator air operating range.
f. Adjustable set pressure range.
g. Manufacturer's model and part number or identification.
h. Manufacturer's drawing numbers, assembly, and component identification.
i. Applicable manual number.
j. Space for nine-digit component identification number.

## 4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are listed in table III and are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

TABLE III. Examination and inspections.

| Characteristic | First article | Conformance | Requirement | Verification |
|---|---|---|---|---|
| Operation | X | | 3.4.8 and 3.4.10 | 4.2.2 |
| Flow | X | | 3.4.6 | 4.2.3 |
| Shock | X | | 3.4.7 | 4.2.4 |
| Vibration | X | | 3.4.7 | 4.2.4 |
| Post-shock operation | X | | 3.4.7 | 4.2.2 |
| Valve configuration | | X | 3.2.1 and 3.3 | 4.3.1 |
| Interchangeability | | X | 3.3.1.3 | 4.3.1 |
| Accessibility | X | | 3.3.1.4 | 4.3.1 |
| Maintainability | X | | 3.3.1.5 | 4.3.1 |
| End preparation | | X | 3.3.2.2 | 4.3.1 |
| Seat leakage | | X | 3.4.3 | 4.3.4 |
| Diaphragm case leakage | | X | 3.4.5 | 4.3.5 |

4.2 First article inspection. First article inspection shall be performed on air-diaphragm-operated control valves and their associated instrumentation when a first article sample is required (see 3.1). This inspection shall include the examination of 4.2.1 and the tests of 4.2.2 through 4.2.4.

4.2.1 Examination. The valve shall be disassembled and visually and dimensionally examined to determine conformance to the requirements of this specification and drawings. Particular emphasis shall be placed on the dimensions, finishes, and condition of the guiding and seating surfaces. In lieu of disassembly for examination, valves shall be visually and dimensionally examined prior to assembly.



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4.2.2 Operational tests. The valve shall be operationally tested as follows:

a. The valve shall be disassembled and subjected to operating pressure and temperature. Stem leakage shall not exceed that specified in 3.4.8. Tapping the valve, or other means to overcome sticking during the test, shall not be permitted.

b. Hysteresis shall be tested at 10, 25, 50, 75, and 90 percent of stroke. A signal change of 0.35 $lb/in^2$ in either direction shall cause valve movement at each position. A hysteresis loop shall be obtained and the maximum difference in valve position between increasing and decreasing pressures shall not exceed 2 percent of stroke.

c. Linearity of travel shall be tested at 0, 25, 50, 75, and 100 percent of stroke. The relationship between air pressure and stem travel shall be linear as specified in 3.4.10.

d. If a valve positioner is supplied, it shall be bypassed for the above tests.

Where the specified operating conditions exceed those available at the manufacturer's facilities, consideration may be given to allowing these tests to be conducted at the closest available operating conditions, provided that the packing's sealing and hysteresis characteristics can be shown to meet the above criteria at the specified operating temperature and pressure conditions.

4.2.3 Flow tests. A flow test shall be conducted to determine the valve $C_v$ and the flow characteristics. Flow rating shall be as specified (see 6.2), and the flow characteristics shall fall within the limits defined (see 3.4.6). This test may be waived where the manufacturer supplies verifiable test evidence based on previous testing that the valve meets these requirements.

4.2.4 Shock and vibration tests. When specified (see 6.2), valves shall be subjected to shock testing in accordance with MIL-DTL-901, grade A, class I, and MIL-STD-798, and the vibration tests specified in MIL-STD-167-1, type I. During and subsequent to these conditions, the equipment shall not fail to perform its principal function, and the deviation in the instrument output pressure signal shall not exceed 3 percent of its output pressure range. Any instrumentation intended to be valve-mounted should be so mounted during shock and vibration testing. Post-shock operational tests shall be as specified in 4.2.2 and deviations shall not exceed the values listed by more than 50 percent. Extension of previous shock and vibration tests may be considered where changes involve such aspects as the substitution of an inner valve design or a reverse acting for a direct acting actuator, or vice versa. Previous acceptance of a body subassembly-actuator combination may extend to the same body subassembly in combination with a smaller actuator, provided that smaller actuator has passed shock and vibration testing.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.3.1 and the tests of 4.3.2 through 4.3.5 on samples selected in accordance with 4.3.a and 4.3.b. This inspection shall include the examinations and tests specified in table III.

a. Sampling for examination. A random sample of valves shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and Acceptance Quality Limit (AQL) specified (see 6.2).

b. Sampling for tests. A random sample of valves shall be selected from each lot offered to the Government in accordance with ASNI/ASQ Z1.4 at inspection level II and AQL specified (see 6.2).

4.3.1 Examination. Each sampled valve shall be visually examined to verify valve configuration (see 1.2) and compliance with the requirements of 3.3.1.3, 3.3.1.4, 3.3.1.5, and 3.3.2.2.

4.3.2 Nondestructive tests. Each sampled valve shall undergo radiographic, magnetic particle, and dye penetrant tests in accordance with S9074-AR-GIB-010/278.

4.3.3 Hydrostatic tests. Each valve shall be hydrostatically tested as follows prior to any painting of the valve body:

a. Composition A, B, and D valves: in accordance with ASME B16.5.

b. Non-ferrous valves: two times the nominal pressure rating.



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4.3.4 Seat leakage tests. Tests for seat leakage shall be conducted on each sampled valve as follows:

a. Valves with a pressure of 3 to 15 lb/in$^2$ gauge: range springs shall be tested with a pressure of 0 or 18 lb/in$^2$ gauge on the diaphragm.

b. Valves with a pressure of 6 to 30 lb/in$^2$ gauge: range springs shall be tested with a pressure of 0 or 36 lb/in$^2$ gauge on the diaphragm.

c. Valves with other ranges shall be tested with zero or the normal supply pressure on the diaphragm.

The line pressure during this test shall be the working pressure and shall be applied in the direction of flow. Maximum allowable leakage shall be as specified in 3.4.3. During this test, a check shall be made for external leakage.

4.3.5 Diaphragm case leakage tests. A leakage test shall be made on each sampled valve with air at the pressure of 50 lb/in$^2$ gauge after complete assembly. Detection shall be by soap bubble solution. There shall be no visible leakage over a 5-minute period.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The air-diaphragm-operated control valves and their associated instrumentation are intended for throttling and on-off liquid service on Naval vessels. Current stock of valves may be used until depleted; future requisitions for these valves should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and composition required (see 1.2).
c. Rating required (see 1.3).
d Trim material required (see 1.4, 3.2.1, and 3.3).
e. Trim size and flange size required (see 1.4 and 3.3).
f. When first article is required (see 3.1).
g. Whether valve is to be used for potable water (see 3.2).
h. Internal trim materials, when materials other than those listed are required (see 3.2.1).
i. Requirements for special tools (see 3.3.1.5).
j. Whether valves should be other than of a basic globe configuration (see 3.3.2.2).
k. Whether quick change trim is required (see 3.3.2.10.2).
l. Whether hand wheel for manual override is required, and if required, its location (see 3.3.3.9).
m. Whether pneumatic instrumentation is required to be furnished (see 3.3.4).
n. Required capacity of valve (see 3.4.1).
o. Flow characteristics required (see 3.4.6 and 4.2.3).
p. Whether mechanical shock and vibration testing are required (see 4.2.4).
q. AQL of 1.0 (see 4.3).



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r. Packaging requirements (see 5.1).

s. Documentation (see 6.4).

t. Service conditions:

(1) Fluid and range of temperatures.

(2) Minimum and maximum flow rates required.

(3) Minimum and maximum inlet and outlet pressure.

(4) Rate of load change.

(5) Control function required.

u. Brief description of application and any special performance or construction requirements.

6.3 Definitions. The definitions of 6.3.1 through 6.3.14 are applicable to this specification.

6.3.1 Air-to-close, normally open. A control valve, actuator plus body subassembly, is of air-to-close construction if an increase in pressure to the actuator causes the plug to move toward the seated position.

6.3.2 Air-to-open, normally closed. A control valve, actuator plus body subassembly, is of air-to-open construction if an increase in pressure to the actuator causes the plug to move away from the seated position.

6.3.3 Constant bleed instrument. An instrument or instrument system that continually exhausts or bleeds a certain amount of the control air to atmosphere, even when the control valve actuator pressure is being maintained at a constant value.

6.3.4 Direct acting actuator. An increase in the pressure to the actuator results in a downward stroke of the stem.

6.3.5 Direct acting control pilot or transmitter. The output signal increases as the measured variable or input signal increases.

6.3.6 Modified linear. A flow characteristic that is a compromise between equal percentage and linear, where the valve exhibits a linear characteristic during the middle portion of the lift and a decreased sensitivity at the lower and upper portions of the list.

6.3.7 Flow coefficient ($C_v$). A basic capacity rating for valves that relates flow rate to the inlet and outlet pressure for a particular fluid. For water service, defined as the number of gallons per minute (gal/min) of 60 °F water that will flow through the valve with a 1-lb/in$^2$ pressure drop across the valve.

6.3.8 Instrumentation. In this specification, refers to any pneumatic instrumentation, control pilots, transmitters, relays, selectors, positioners, instrument air reducing valves, and strainers required for operation of the control valve in the system.

6.3.9 Intermittent bleed instrument. An instrument or instrument system that only exhausts or bleeds control air to atmosphere when it is necessary to decrease the control valve actuator pressure. Under steady load conditions where there is no change in actuator pressure required, both the supply and vent functions are shut off and there is no loss of control air to atmosphere.

6.3.10 Internal trim. Trim including internal parts of the control valve such as seat rings, plugs, stems, guide bushings, cage, and pistons.

6.3.11 Quick change cage trim. Internal trim consisting of a gasket sealed seat ring held in position by a cage, which may be either separate from or integral with the seat ring, which is in turn held in position by either the bonnet or bottom flange. This design should permit the rapid replacement of all internal trim by avoiding the use of any threads located within the valve body, such as seat ring threads.

6.3.12 Rangeability. A measure of the usable range of a control valve and defined as the ratio of the maximum to the minimum controllable $C_v$. This maximum and minimum controllable $C_v$ establishes the throttling range over which a given control characteristic can be maintained and within which the valve can perform a useful throttling function.



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6.3.13 Reverse acting actuator. An increase in the pressure to the actuator results in an upward stroke of the stem.

6.3.14 Reverse acting control pilot or transmitter. The output signal decreases as the measured variable or input signal increases.

6.4 Documentation. The following is a list of documentation that may be requested as part of the acquisition of any valve under this specification (see 6.2):

a. Engineering drawings.
b. Master drawings.
c. Materials listing.
d. Installation drawings.
e. Technical manual.
f. Test reports.
g. Certifications.
h. On board repair parts list.

6.5 Subject term (key word) listing.

Cobalt alloy
Diaphragm actuator
Flow coefficient
Monel
Valve, on-off
Valve, throttling

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

## CONCLUDING MATERIAL

Review activity:
DLA – SS

Preparing activity:
Navy – SH
(Project 4820-2018-007)

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